Siddhant Bansal

PhD Candidate in Computer Science

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Profile

Six years of research experience (MSc+PhD) in Video Understanding. Third year PhD researcher in Computer Vision at the University of Bristol, supervised by Prof. Dima Damen. My research focuses on understanding hand-object interactions in two- and three-dimensions, particularly in egocentric setting. I am also a guest scientist at Max Planck Institute for Intelligent Systems working with Prof. Michael Black. Previously, supervised by Prof. C V Jawahar as part of my 3-years masters in video understanding.

Expertise

Egocentric / Embodied Video Understanding

- Worked on **3D-aware egocentric perception** with interaction-timeline reconstruction and geometry-constrained reasoning over object's pose across time (useful for manipulation understanding and state tracking). [**P1**]
- Contributed to 3D-grounded egocentric datasets to enable **3D-consistent evaluation** beyond 2D metrics. [**P2**]
- Proposed learning frameworks for **procedure learning** from untrimmed egocentric video using self-supervised and graph-based modelling of temporal structure. [**P6, P7**]

Generative AI & Multimodal Foundation Models (VLMs / MLLMs)

- Built and benchmarked multimodal evaluation suites for video-language foundation models, targeting spatio-temporal reasoning, activity understanding, and grounding in egocentric video. [**P2, P3, P5, P8**]
- Released large-scale instruction-style QA supervision for egocentric interaction understanding (millions of QA pairs) along with VLM to ground in egocentric view. [**P3**]

Datasets, Benchmarks & Evaluation at Scale (Ego4D, Ego-Exo4D, HD-EPIC)

- Contributed to **large-scale public datasets** and maintained benchmark + baseline pipelines to drive adoption across academia/industry. [**P5, P8**]
- Built **task definitions + metrics** that make model progress measurable (procedure learning, HOI reference/QA, multi-view understanding), accelerating iteration for foundation-model training/eval. [**P2, P3, P7, P8**]

Publications & Preprints [P]

*(Author order as in papers; * indicates equal contribution)*

1. **Reconstructing Objects along Hand Interaction Timelines in Egocentric Video.** Zhifan Zhu, **Siddhant Bansal**, Shashank Tripathi, Dima Damen. *arXiv* 2025. paper | project | code | results video
2. **HD-EPIC: A Highly-Detailed Egocentric Video Dataset.** Toby Perrett* *et al.* (incl. **Siddhant Bansal***). **CVPR 2025**. paper | project | code+data
3. **HOI-Ref: Hand-Object Interaction Referral in Egocentric Vision. Siddhant Bansal**, Michael Wray, Dima Damen. *arXiv* 2024. paper | project | code+data
4. **An Outlook into the Future of Egocentric Vision.** Chiara Plizzari*, Gabriele Goletto*, Antonino Furnari*, **Siddhant Bansal***, Francesco Ragusa*, Giovanni M. Farinella, Dima Damen, Tatiana Tommasi. **IJCV 2024**. paper
5. **Ego-Exo4D: Understanding Skilled Human Activity from First- and Third-Person Perspectives.** Kristen Grauman *et al.* (incl. **Siddhant Bansal***). **CVPR 2024**. paper | project | code
6. **United We Stand, Divided We Fall: UnityGraph for Unsupervised Procedure Learning from Videos. Siddhant Bansal**, Chetan Arora, C. V. Jawahar. **WACV 2024**. paper | project
7. **My View is the Best View: Procedure Learning from Egocentric Videos. Siddhant Bansal**, Chetan Arora, C. V. Jawahar. **ECCV 2022**. paper | project | code
8. **Ego4D: Around the World in 3,000 Hours of Egocentric Video.** Kristen Grauman *et al.* (incl. **Siddhant Bansal**). **CVPR 2022 (Oral; Best paper finalist)**. paper | project | code

9. **Improving Word Recognition using Multiple Hypotheses and Deep Embeddings. Siddhant Bansal**, Praveen Krishnan, C. V. Jawahar. **ICPR 2020**. paper | project | code

10. **Fused Text Recogniser and Deep Embeddings Improve Word Recognition and Retrieval. Siddhant Bansal**, Praveen Krishnan, C. V. Jawahar. **DAS 2020 (Oral)**. paper | project | code

Education

University of Bristol, United Kingdom *Apr 2023 – Present*
PhD in Computer Science Advisor: Prof. Dima Damen

IIIT Hyderabad (CVIT), India *Jan 2020 – Mar 2023*
MS (by Research) in Computer Science CGPA: 9.2/10.0
Thesis: I-Do, You-Learn: Techniques for Unsupervised Procedure Learning using Egocentric Videos (PDF)
Advisors: Prof. C. V. Jawahar; Prof. Chetan Arora

Vishwakarma Government Engineering College, Ahmedabad, India *Jun 2015 – Jul 2019*
BE in Electronics and Communication CPI: 8.01/10.0
Capstone: Automatic Garbage Detection and Collection (Top 4% SSIP projects; full government funding)

Research Experience

PhD Researcher, University of Bristol *Apr 2023 – Present*
Advisor: Prof. Dima Damen

- **HD-EPIC (CVPR 2025):** contributed to a highly-detailed kitchen egocentric dataset with **41 hours** of unscripted recordings, **3D digital twins**, dense multi-modal annotations, and a **26.6K** VQA benchmark for evaluating VLMs.
- **HOI-Ref (VLM for hand–object interaction referral):** proposed a novel referral task for egocentric images; curated HOI-QA with 3.9M QA pairs; trained VLM4HOI (Llama2-7B based) demonstrating strong gains from egocentric fine-tuning.
- **Ego-Exo4D (CVPR 2024):** contributed to a large consortium dataset for skilled activity understanding with synchronised ego/exo video and multi-modal sensors; supported benchmarks and open-source release practices for reproducible research.
- Contributed to **ROHIT** (Reconstructing Objects along Hand Interaction Timelines) by modeling temporal constraints over object hand-interaction timelines and evaluating constrained optimisation-based pose propagation on egocentric datasets.

Guest Scientist, Max Planck Institute for Intelligent Systems (MPI-IS), Tübingen *Sep 2025 – Present*
Host: Prof. Michael J. Black

- Working on hand-object 3D reconstruction (*under review*).

MS Researcher, Center for Visual Information Technology (CVIT), IIIT Hyderabad *Jan 2020 – Mar 2023*
Advisors: Prof. C. V. Jawahar; Prof. Chetan Arora

- **EgoProceL (ECCV 2022):** created a large-scale procedure-learning dataset (**62 hours**, 130 subjects, 16 tasks) and proposed **Correspond-and-Cut** self-supervised learning using key-step temporal correspondences.
- **UnityGraph (WACV 2024):** proposed a graph-based procedure learning framework that leverages intra-/inter-video context; improved performance on ego procedure learning (+3.6% on EgoProceL).
- **Ego4D (CVPR 2022 Oral):** contributed to the Hands & Objects benchmark effort; built baseline training/evaluation pipelines and supported data governance workflows (IRB, consent, de-identification).
- **Word recognition & multilingual OCR (ICPR/DAS 2020):** improved word recognition via multiple hypotheses + deep embeddings; built OCR for Hindi/Tamil/Telugu and explored semi-supervised CRNN training with CTC loss.

Research Intern, IIT Gandhinagar *Mar 2019 – Aug 2019*
Advisor: Prof. Shanmuganathan Raman

- Cultural heritage 3D modeling: point-cloud acquisition and alignment (ICP) and point-cloud completion with autoencoders.

AI Intern, Meditab Software Inc. (India) *Sep 2018 – Mar 2019*

- Evolutionary layout optimization for robotic systems (reported 75% reduction in travel time).

AI Research Intern, Bennett University, India *Jun 2018 – Jul 2018*

- Collected a dataset of footprints (180 volunteers) and trained a deep model for footprint-based human classification.

Data Analyst Intern, Bioscan Research, Ahmedabad, India *Jun 2018 – Jul 2018*

- Built a Python+SQLite patient-tracking tool and signal extraction utilities for an onsite intracranial-bleeding scanning tool.

Academic Service & Leadership

- **Workshop organiser:** EgoVis @ CVPR 2026; EgoVis @ CVPR 2025; EgoVis @ CVPR 2024; Ego4D+EPIC Workshop @ CVPR 2023; Ego4D Workshop @ ECCV 2022.
- **Reviewer:** CVPR2022–2026; ICCV2023&2025 (**Outstanding Reviewer**); ECCV2022&2024; WACV2023–2024; ICRA2025.
- **Journal reviewer:** IEEE TPAMI; IJCV; CVIU.

Awards & Funding

- **Outstanding Reviewer**, ICCV 2025.
- **Google Summer of Code (GSoC)**, 2022: Enhancing DBpedia with image-based querying.
- **Standing Committee Admission** award for MS by Research at IIIT Hyderabad.
- **Student Startup and Innovation Policy**: Top 4% project selected for full funding (Automatic Garbage Collection).
- Supported by a charitable donation to University of Bristol from Meta.

Technical Skills

Programming:	Python (primary), SQL, Bash
DL/ML:	PyTorch; HuggingFace/Transformers; TensorFlow/Keras (prior); training & evaluation of VLMs/LLMs;
CV/Video:	OpenCV; video understanding; HOI reasoning; 3D geometry (point clouds, ICP)
Tooling:	Git, Linux, Conda, LaTeX; Docker; multi-GPU training (`torchrun`)